AVANTIKA RATTAN

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SUMMARY

Robotics engineer and M.S. Robotics student at the University of Michigan (CURLY Lab, advisor Prof. Maani Ghaffari, graduating May 2027) working across state estimation, perception, and deep learning on real robots: Fetch, Husky UGV, an autonomous surface vessel, and a 6-DOF arm. Current Graduate Student Instructor for ROB 330. GPA 3.978/4.00. Co-authored paper accepted at CoRL 2026; TartanIMU Challenge 8th overall with an award and IROS 2026 workshop presentation; first-author IEEE Access 2025 review.

EDUCATION

M.S. Robotics, University of Michigan, Ann Arbor – Expected 05/2027, GPA 3.978/4.00
Courses (grade): Robotic Systems Laboratory, ROB 550 (A+); Mobile Robotics, ROB 530 (A+); Self Driving Cars, ROB 535 (A); Deep Learning for Robot Perception, ROB 599 (A); Math for Robotics, ROB 501 (B+). In progress: 3D Robot Perception, ROB 599; Natural Language Processing, CSE 595
B.Tech, Mechatronics and Automation, Vellore Institute of Technology – 05/2024, CGPA 9.32/10, Ranked 3rd

TEACHING EXPERIENCE

Graduate Student Instructor, ROB 330 Localization, Mapping, and Navigation, University of Michigan – Fall 2026

- Support labs on the MBot platform (particle filter localization and path planning) and hold weekly office hours.

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant, CURLY Lab, University of Michigan – May 2026 to Present

- Built a camera and LiDAR perception pipeline for the Nautilus autonomous surface vehicle with Honda Research Institute: assembled a 13,083-image dataset with SAM2-assisted annotation, trained YOLO11 segmentation (Box mAP50 0.993, Mask mAP50 0.981), synchronized the sensors in a 120 ms window, and fused detections with LiDAR using DBSCAN, RANSAC, and PCA geometry (3.15 m mean close-range 2D RMSE).
- Developed SO(3)-equivariant inertial odometry networks and ran them in a real-time ROS2 inference pipeline on a Fetch robot and a Husky UGV, with robot-specific configurations, tested in controlled hardware experiments (GINIO, accepted at CoRL 2026; 16.5 times lower trajectory error than a standard ResNet on the Fetch).
- Diagnosed and fixed a full-disk failure on an NVIDIA Jetson Orin by redirecting storage from the 57 GB eMMC to NVMe, freeing about 28 GB.

Research Collaborator, Honda Research Institute (via CURLY Lab) – May 2026 to July 2026

- Traced bad multi-sensor (GPS, IMU, camera, LiDAR) trailer tracking results to a coordinate frame (TF tree) fault and delivered a root-cause failure analysis in a technical report.

Robotics and Research Engineer Intern, Lacuum AI – June 2026 to August 2026

- Built an autonomy stack for an autonomous boat in VRX/Gazebo simulation: GPS and IMU localization, camera obstacle mapping (ConvBKI), a global map from Google Maps and OpenStreetMap satellite imagery, and A* path planning for a full-lake survey.
- Validated GPS pose tracking on the real boat while it was moving (towed behind an ATV, no camera or LiDAR for that test), matched against satellite imagery of the pond in Rviz.

PROJECT PORTFOLIO (code: github.com/Avantikarattan6134, models: huggingface.co/arattan7272)

1. GINIO: SO(3)-Equivariant Neural Inertial Odometry (CURLY Lab; CoRL 2026, fourth author)

- **Problem:** Standard neural inertial odometry networks degrade sharply when the IMU is mounted in a different orientation than in training (up to 81 times worse trajectory error in our remounting tests).
- **Architecture:** SO(3)-equivariant Vector Neuron (VN) and Lie Neuron (LN) ResNets that take gravity-aligned IMU windows and predict displacement, so a rotation of the input produces a matching rotation of the output.
- **My work:** Ran the IMU remounting experiments to collect equivariance validation data on a Fetch robot (BNO055 IMU; test cases with a 90 degree roll about x and a 90 degree rotation about y) and a Husky UGV (LORD Parker 3DMGX5 IMUs at 1000 Hz). Processed the data into a 17-column format at 100 Hz (timestamps, world-frame gyro and accelerometer, orientation quaternion, position, velocity), with a preprocessing chain that rotates raw IMU readings into the world frame, removes bias, and subtracts gravity. Built the real-time ROS2 inference pipeline for

the LN-ResNet, VN-ResNet, and standard ResNet models with robot-specific YAML configurations. I also worked on the peer-review rebuttal experiments.

- **Result:** 16.5 times lower trajectory error than a standard ResNet on the Fetch (2.018 m vs 76.389 m); validated on the Husky UGV.

2. ENIO: TartanIMU Challenge (Kaggle, IROS 2026 workshop; 8th overall, challenge award)

- **Problem:** Estimate motion from IMU data across wheeled, legged, handheld, and aerial platforms, with no platform label given at test time.
- **Architecture:** One 5.38M-parameter equivariant network (LN-ResNet backbone) with a cross-window BiLSTM that shares information between consecutive windows. A single model serves all platforms.
- **Method:** Window-offset sampling to create about 200 times more training data from the same recordings; about 50 controlled experiments with negative results kept; every claim checked on held-out data; about 70 minutes of training on one GPU.
- **Result:** 0.25650 vs the organizers' 0.538 baseline (lower is better), 8th overall. Presented the method at the workshop on Oct 4, 2026, and was invited to contribute to the IMU Foundation Model white paper. Model released at huggingface.co/arattan7272/tartanimu-lnresnet.

3. Nautilus ASV Perception and SLAM (Honda Research Institute, CURLY Lab)

- **Perception pipeline:** YOLO11 instance segmentation on 2880 by 2880 images, then synchronization with the LiDAR point cloud in a 120 ms window, projection of filtered LiDAR points onto the segmented regions, aggregation over 25 frames, DBSCAN clustering, RANSAC plane fitting, and a two-pass PCA that estimates object position and orientation with a distance-based uncertainty.
- **SLAM pipeline (in parallel):** DLIO estimates motion between LiDAR scans; a GTSAM pose graph combines odometry, landmark, and loop closure constraints into a drift-free 3D map.
- **Data and training:** SAM2 in video-propagation mode accelerated annotation; combined Florida and Japan dataset of 13,083 images; 200 epochs at 2880 resolution on an RTX 4090. Model-size sweeps on the Great Lakes cluster (A40 nodes) found the nano segmentation model beat larger variants.
- **Results:** Box mAP50 0.993, Mask mAP50 0.981; 3.15 m mean close-range 2D RMSE over three approach scenarios on the Florida bags. Written up in a mid-project technical report. DLIO failed on the Florida bags because of a missing TF tree, which was documented as a finding.
- **Embedded systems:** Fixed a full-disk failure on the NVIDIA Jetson Orin AGX (57 GB eMMC at 100 percent) by backing up, moving Docker, conda, pip cache, and ROS data to an NVMe SSD and symlinking them back, freeing about 28 GB.

4. Honda Trailer Tracking Failure Analysis (Honda Research Institute)

- Multi-sensor tracking (GPS, IMU, camera, LiDAR) of a trailer gave bad results. I traced the root cause to a coordinate frame (TF tree) fault and delivered a failure analysis in a technical report. OpenCV was used for the vision processing.

5. Autonomous Boat Autonomy Stack (Lacuum AI internship, June to August 2026)

- **Architecture (simulation, VRX/Gazebo):** GPS and IMU localization; camera-based obstacle mapping with ConvBKI (Convolutional Bayesian Kernel Inference); a global map built from Google Maps and OpenStreetMap satellite imagery; A* path planning; an end-to-end autonomous full-lake survey.
- **Real boat:** Only GPS pose tracking has been validated while moving (the boat was towed behind an ATV, with no camera or LiDAR mounted); the track matched satellite imagery of the pond in Rviz. Planning and obstacle mapping ran in simulation.

6. Impact-Aware State Estimation (ROB 530, with Pranav Jain)

- **Architecture:** Right-Invariant Extended Kalman Filter on SE2(3) written in C++, with an LSTM that adapts the noise model. Evaluated on the ROOAD driving dataset.
- **Result:** Trajectory error cut 96 percent (6.326 m to 0.258 m).

7. ROB 550 ArmLab: Manipulation on a 6-DOF Arm

- Built a full manipulation pipeline: forward and inverse kinematics, camera calibration, vision-based object detection and pose estimation, grasping, and path planning, through the final competition checkpoint.

8. Invasive Aquatic Plant Segmentation (autonomous removal)

- Built a dataset of 20 Michigan invasive aquatic plant species from the MSU Extension field guide, pulled through the iNaturalist API with per-species regional bounding boxes (PowerShell scripts). Annotated with SAM2-backed

Smart Polygon in Roboflow and trained YOLO11 instance segmentation.

9. CARLA-PerceptionLab

- Synthetic data pipeline in the CARLA simulator with domain randomization for driving object detection; 0.757 mAP.

10. Vision-Guided Screw Assembly Robot (group project with Kevin Biju Mathew and Aaditya Abhang)

- The robot reads screw requirements with OCR, decides which screw type to pick, picks it, places it at the right spot on a board, and fastens it. Computer vision handles reading and screw identification. Presented with problem, procedure, results, and success measures. [Add your specific results here.]

11. Coursework and Learning Projects

- **Differentiable Particle Filters:** compared five observation models for learned state estimation.
- **BEVUNet:** bird's-eye-view segmentation. **Other models used:** PointNet, ViT, UNet. **Reinforcement learning:** PPO and DQN. **Planning:** MGO-RRT*.

12. In Progress

- **SE(3)-equivariant Vision Transformer** for camera and IMU visual-inertial odometry; no results yet.
- **3D reconstruction and scene understanding:** Gaussian splatting, NeRF, and MinkowskiNet; no results yet.
- **IDEAS Lab drone research:** autonomous drones that map large forests and predict the motion of multiple people; ongoing.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, Bash
Perception: YOLO11 detection and segmentation, SAM2-assisted annotation, OpenCV, LiDAR processing (DBSCAN, RANSAC, PCA), camera calibration, sensor synchronization
Robotics: ROS2, state estimation (EKF, GTSAM pose graphs), LiDAR and camera sensing, VRX/Gazebo, CARLA
Deep Learning and Tools: PyTorch, TensorFlow, YOLO11, SAM2, Docker, Linux, Git, NVIDIA Jetson Orin, Slurm GPU clusters (A40, V100)

AWARDS AND SERVICE

- TartanIMU Challenge award and IROS 2026 workshop presentation (Oct 2026); invited to contribute to the IMU Foundation Model white paper.
- Graduated 3rd in cohort, B.Tech Mechatronics and Automation, Vellore Institute of Technology.

PUBLICATIONS

- GINIO: Geometric Interface for Neural Inertial Odometry. Conference on Robot Learning (CoRL), 2026. Accepted. Fourth author.
- Rattan, A., et al. Monitoring the Condition of Nitrogen-Filled Tires Using Weightless Neural Networks. Automatika, 65(2):523–537, 2024. Co-authored.
- Rattan, A., et al. Quantum Computing for ADAS and Autonomous Vehicles: A Review. IEEE Access, 13:17554–17582, 2025. First author.